D Y PATIL AGRICULTURE & TECHNICAL UNIVERSITY TALSANDE

MCA SANDWICH

PROJECT NAME

Heart Disease Analysis

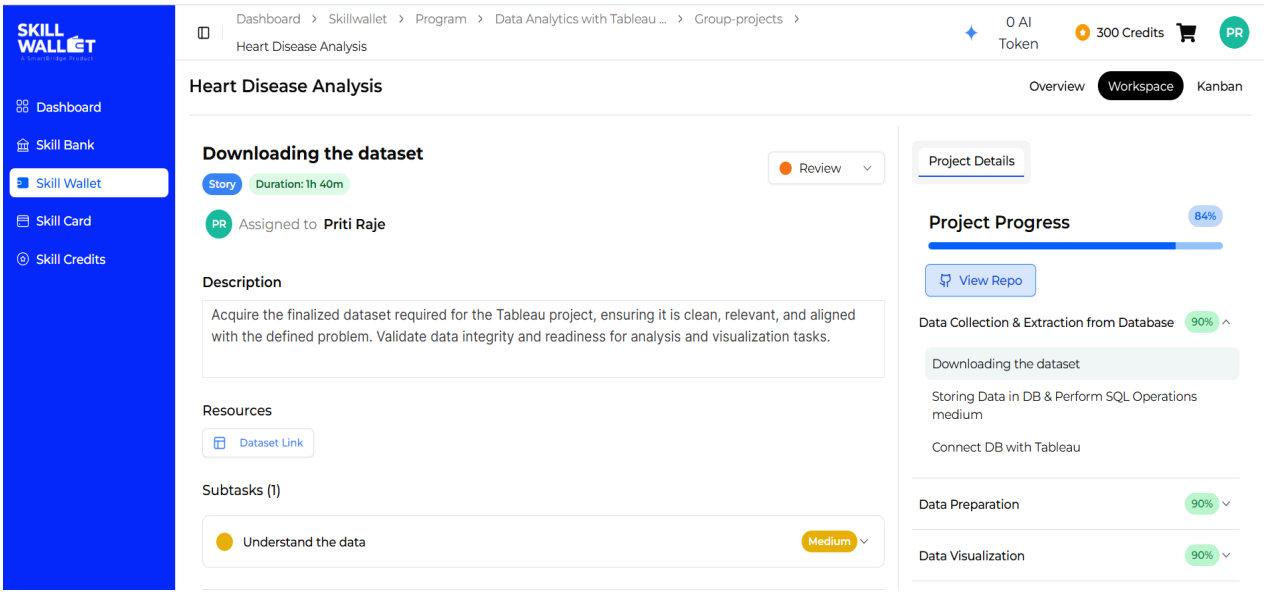
TEAM MEMBERS

1) Priti Raje
2) Sumit Sawant
3) Sandeep Dhole
4) Sanika Jadhav

1) Data Collection & Extraction from Database:

❖ Collect the dataset:

1) Data is Collected from Dataset link Which is provided already so click on **Dataset Link**.



2) From the given dataset link drive opens , then download the dataset
   i.e. **Heart_new2.csv**

❖ Loading the Dataset:

Step 1: Click on "Microsoft Excel" or "Text File" On the Start Page, under Connect, you will see options like:

Microsoft Excel (for .xlsx) Text File (for .csv)
Click the correct option.

Step 2: Select Your File Browse your computer.
Select your dataset file.
Click Open.

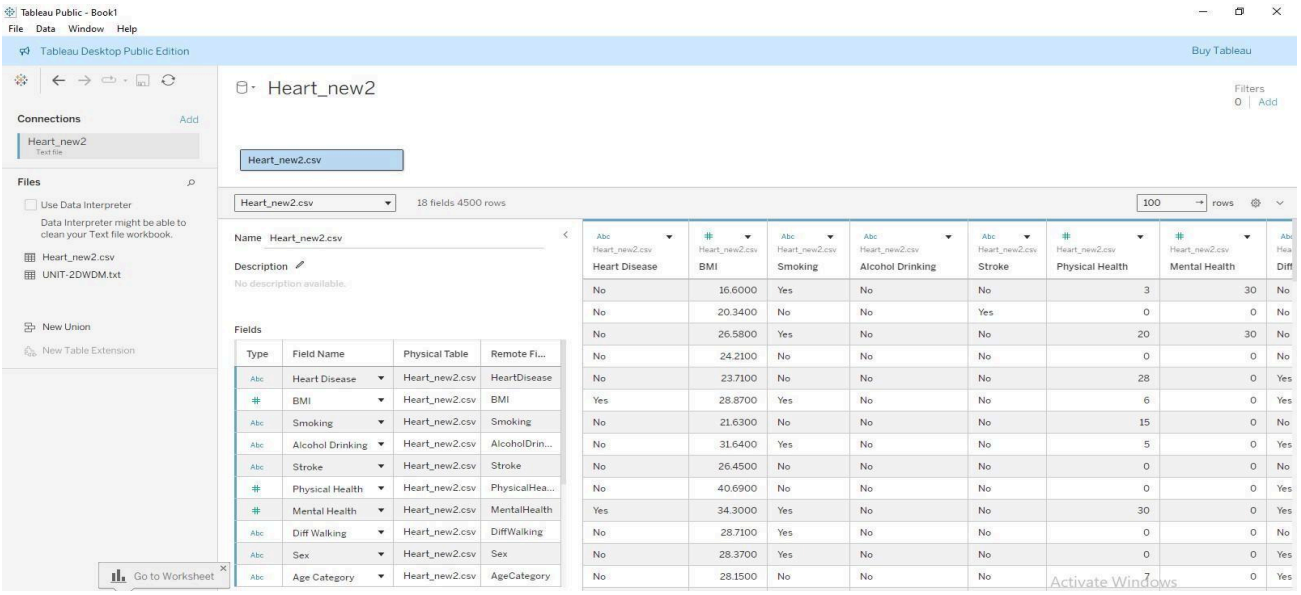
Step 3: Data Source Page

Your sheet names will appear on the left side.
Drag the sheet to the centre area.
Tableau will show a preview of your data.

2) Data Preparation:

❖ Data Cleaning

Preparing the data for visualization involves cleaning the data to remove irrelevant or missing data, transforming the data into a format that can be easily visualized, exploring the data to identify patterns and trends, filtering the data to focus on specific subsets of data, preparing the data for visualization software, and ensuring the data is accurate and complete.



3) Data Visualization:

Data visualization is the process of creating graphical representations of data to help people understand and explore the information. The goal of data visualization is to make complex data sets more accessible, intuitive, and easier to interpret. By using visual elements such as charts, graphs, and maps, data visualizations can help people quickly identify patterns, trends, and outliers in the data.

❖ No of Unique Visualizations:

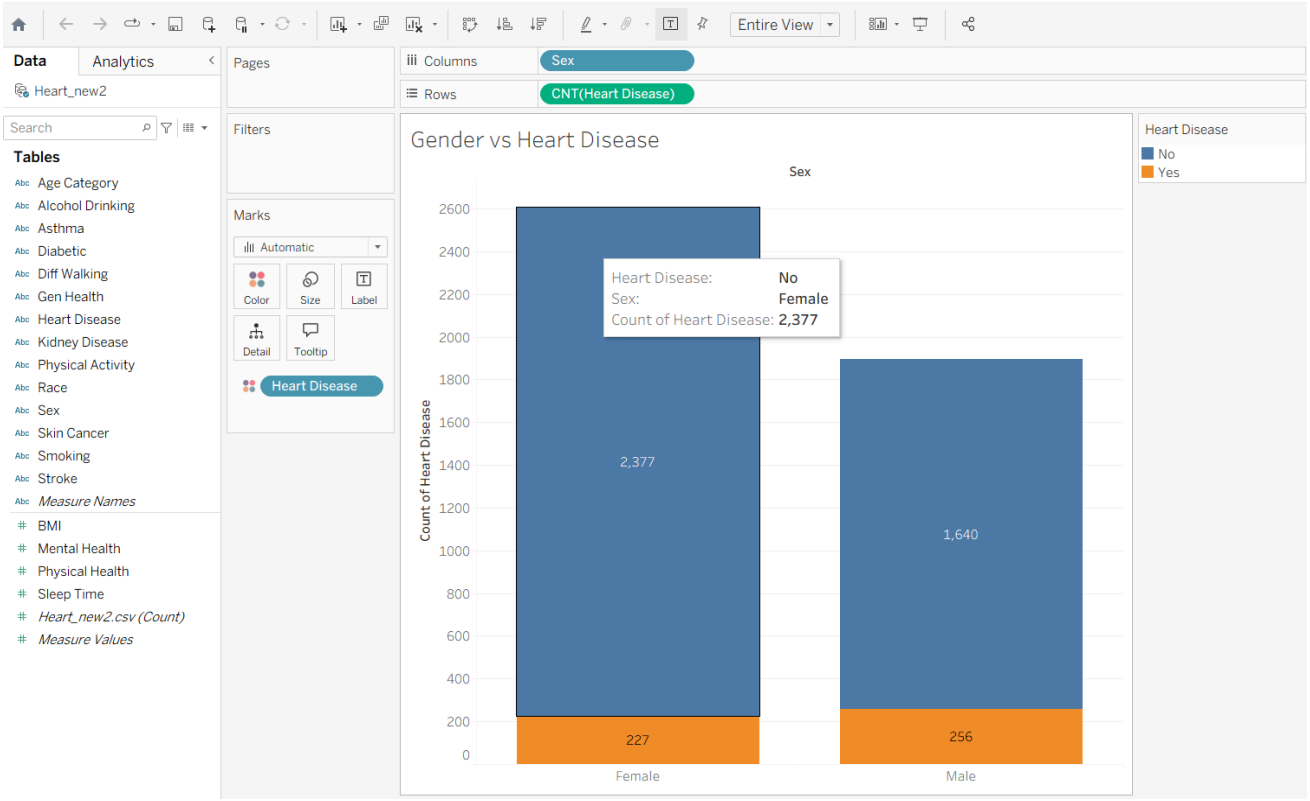
3) Data Visualization:

Data visualization is the process of creating graphical representations of data to help people understand and explore the information. The goal of data visualization is to make complex data sets more accessible, intuitive, and easier to interpret. By using visual elements such as charts, graphs, and maps, data visualizations can help people quickly identify patterns, trends, and outliers in the data.

❖ No of Unique Visualizations:

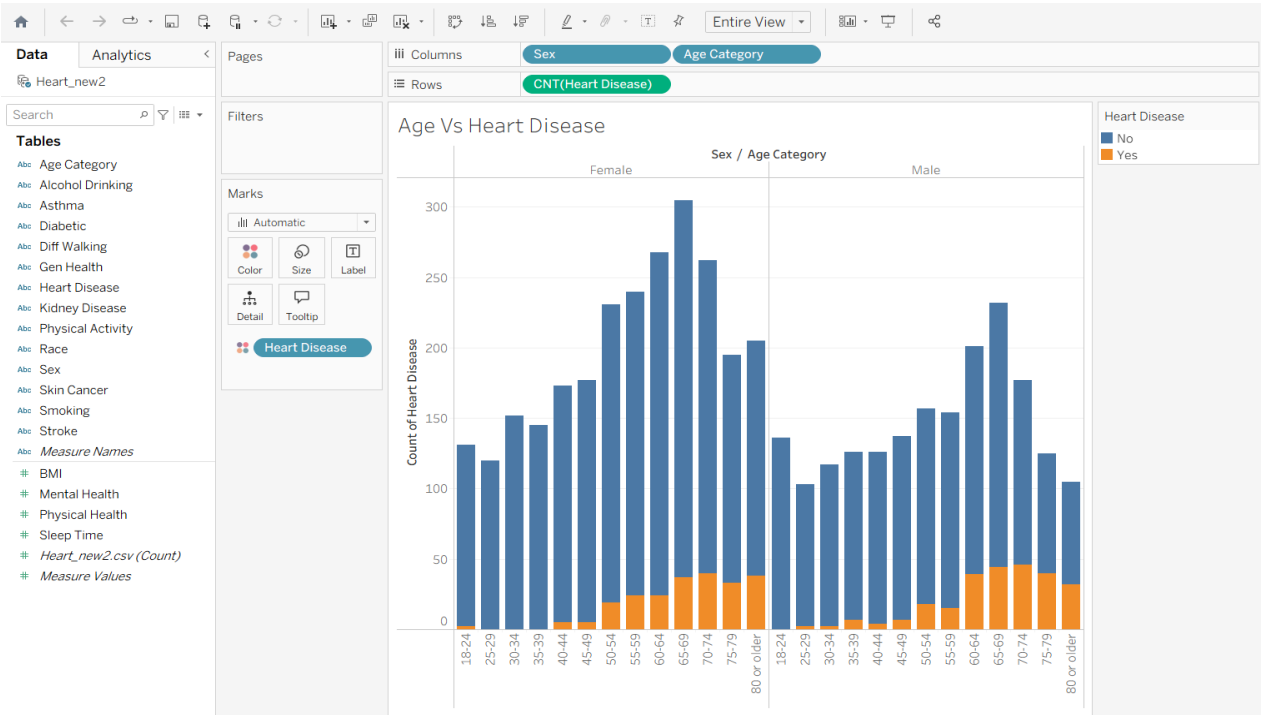
1) Gender wise Heart Disease

a) Drag and Drop Sex into Columns section.
b) Drag and Drop HeartDisease into Rows section.
c) Add HeartDisease into Color from the Marks card.
d) Select Stacked Bar Chart from Show Me panel.
e) Add CNT(HeartDisease) into Label to display count values.



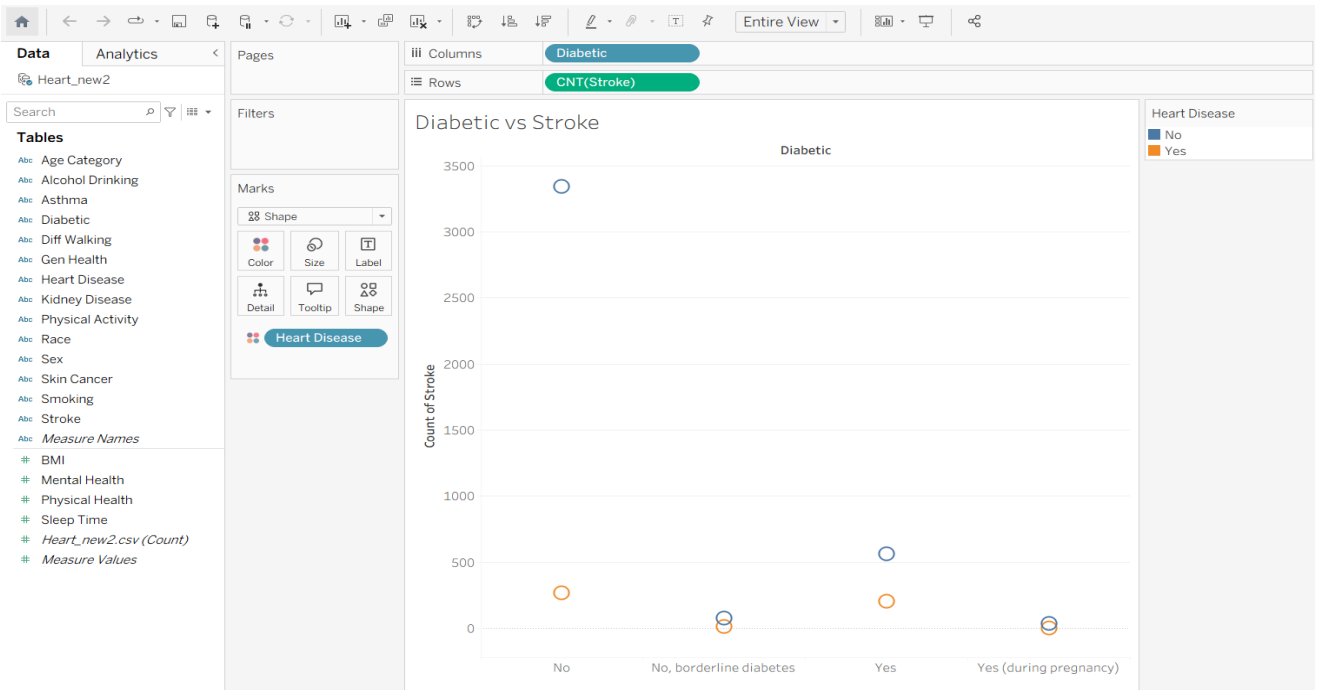
2) Age wise Heart Disease

a) Drag and Drop AgeCategory into Columns section.
b) Drag and Drop CNT(HeartDisease) into Rows section.
c) Add Sex into Color from the Marks card.
d) Select Grouped Bar Chart from Show Me panel.
e) Filter HeartDisease to show both Yes and No values side by side.



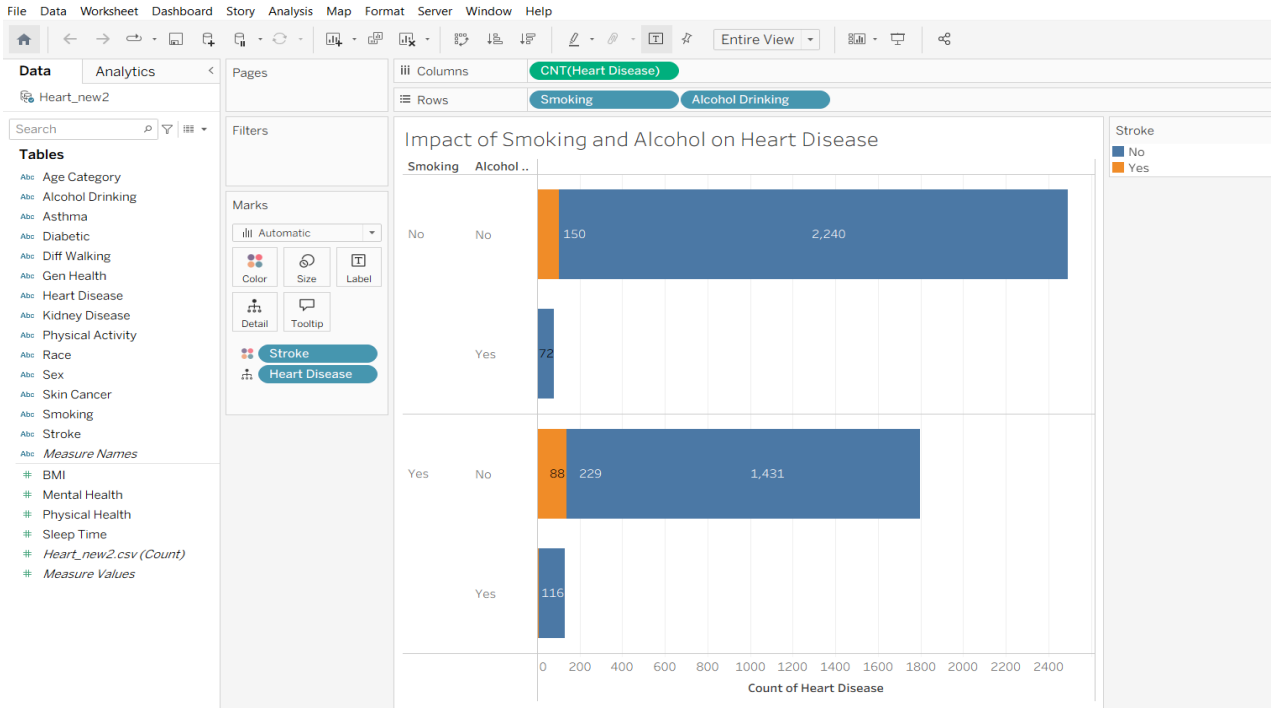
3) The Silent Connection — Diabetics vs Stroke

a) Convert Diabetic to Dimension by right clicking.
b) Drag and Drop Diabetic into Columns section.
c) Drag and Drop CNT(Stroke) into Rows section.
d) Add HeartDisease into Color from the Marks card.
e) Select Scatter Plot from Show Me panel.
f) Add Diabetic into Label to display category names on each point.



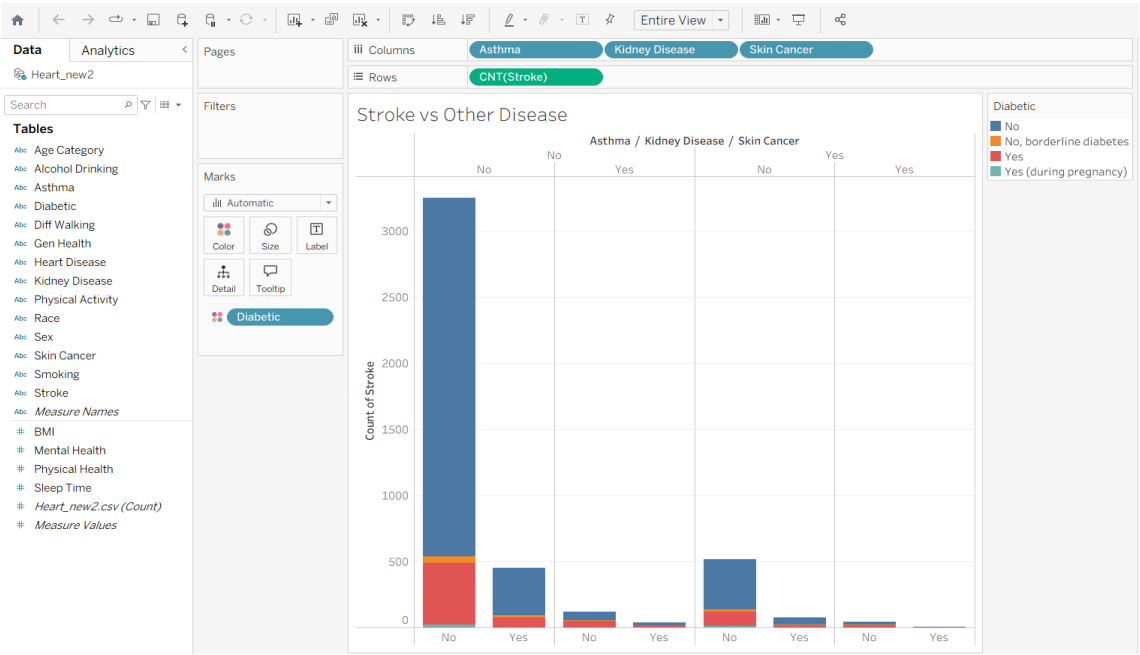
4) Double Trouble — Smoking and Alcohol vs Heart Disease

a) Drag and Drop Smoking into Columns section.
b) Drag and Drop AlcoholDrinking into Columns section alongside Smoking.
c) Drag and Drop CNT(HeartDisease) into Rows section.
d) Add HeartDisease into Color from the Marks card.
e) Select Horizontal Bar Chart from Show Me panel.
f) Add SUM(HeartDisease) into Label to display count values on each bar.



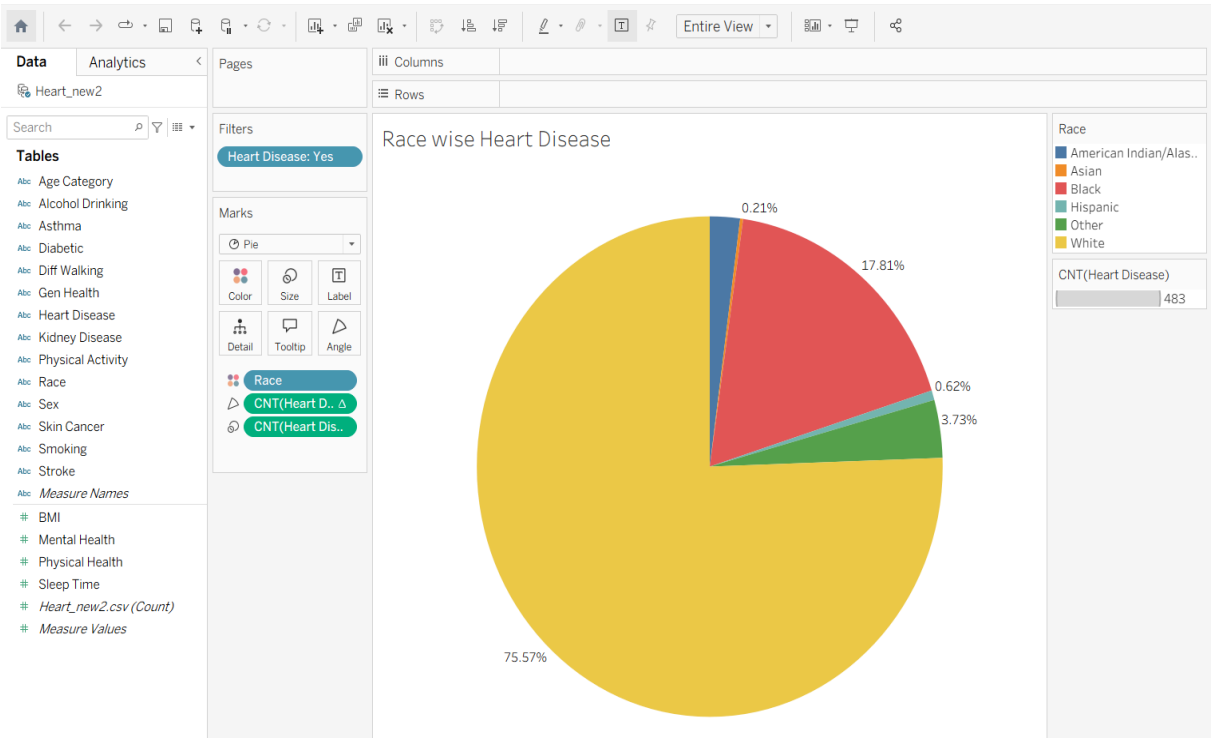
5) When One Disease Brings Another — Stroke vs Other Diseases

a) Drag and Drop Stroke into Columns section.
b) Drag and Drop CNT(HeartDisease) into Rows section.
c) Add Asthma, KidneyDisease, and SkinCancer into the Filters card.
d) Add Stroke into Color from the Marks card.
e) Select Bar Chart from Show Me panel.
f) Add CNT(HeartDisease) into Label to display values on each bar.



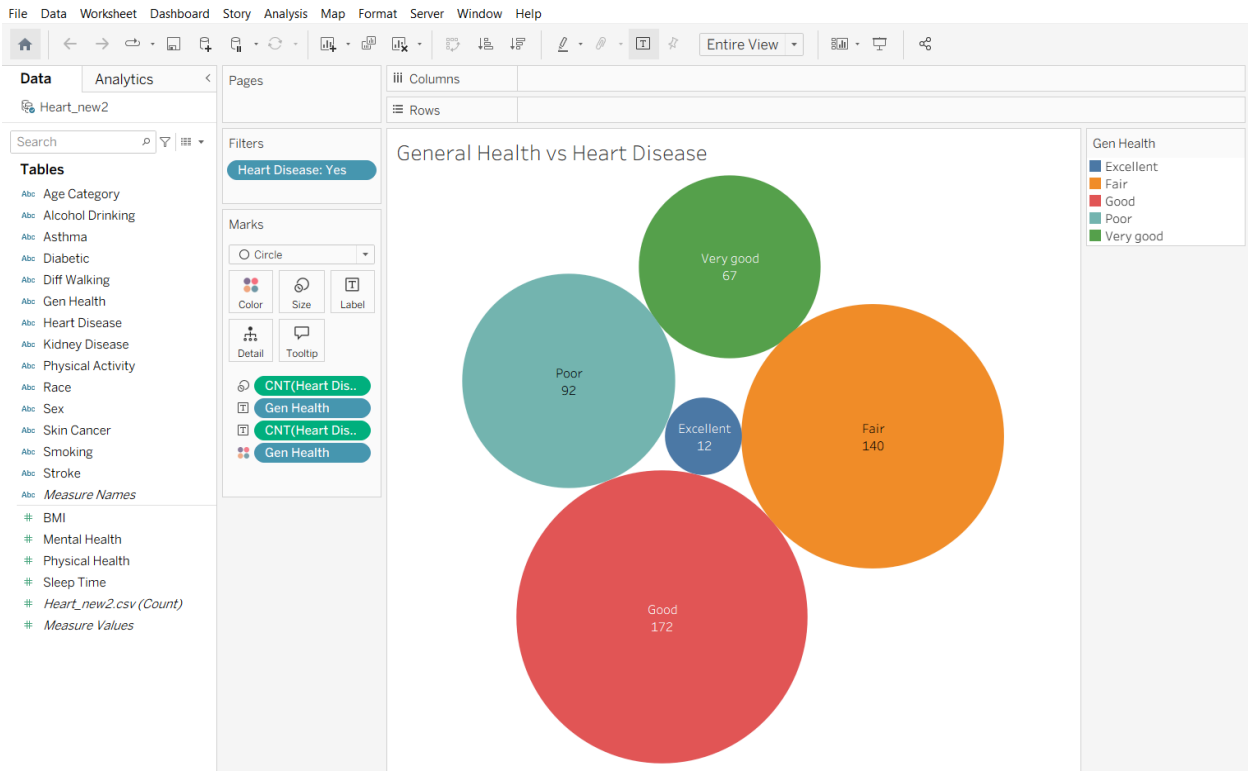
6) A Disease Without Borders — Race wise Heart Disease

a) Drag and Drop Race into the Rows section.
b) Drag and Drop CNT(HeartDisease) into the Angle shelf on the Marks card.
c) Add Race into Color from the Marks card.
d) Select Pie Chart from Show Me panel.
e) Add Race and CNT(HeartDisease) into Label to display category and percentage.
f) Filter HeartDisease to Yes to show only confirmed cases.



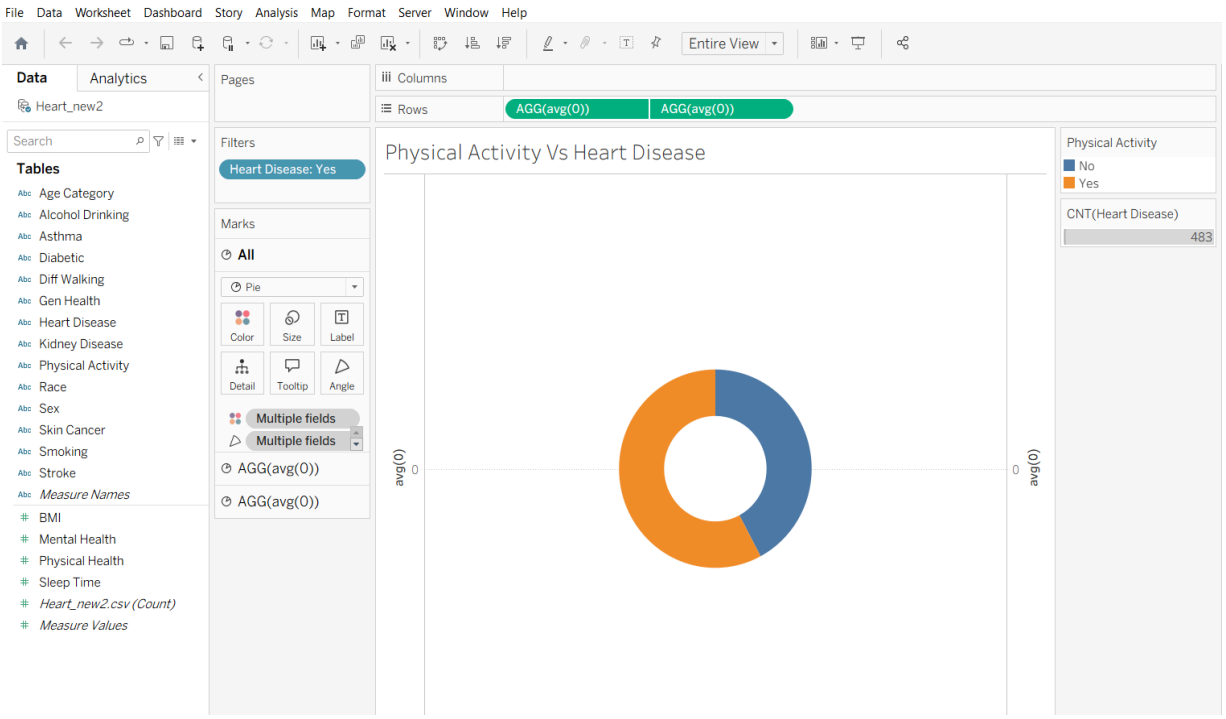
7) Feeling Good Isn't Always Enough — General Health vs Heart Disease

a) Drag and Drop GenHealth into Columns section.
b) Drag and Drop CNT(HeartDisease) into Rows section.
c) Add GenHealth into Color from the Marks card.
d) Add CNT(HeartDisease) into Size from the Marks card.
e) Select Bubble Chart (packed bubbles) from Show Me panel.
f) Add GenHealth and CNT(HeartDisease) into Label to display values on each bubble.



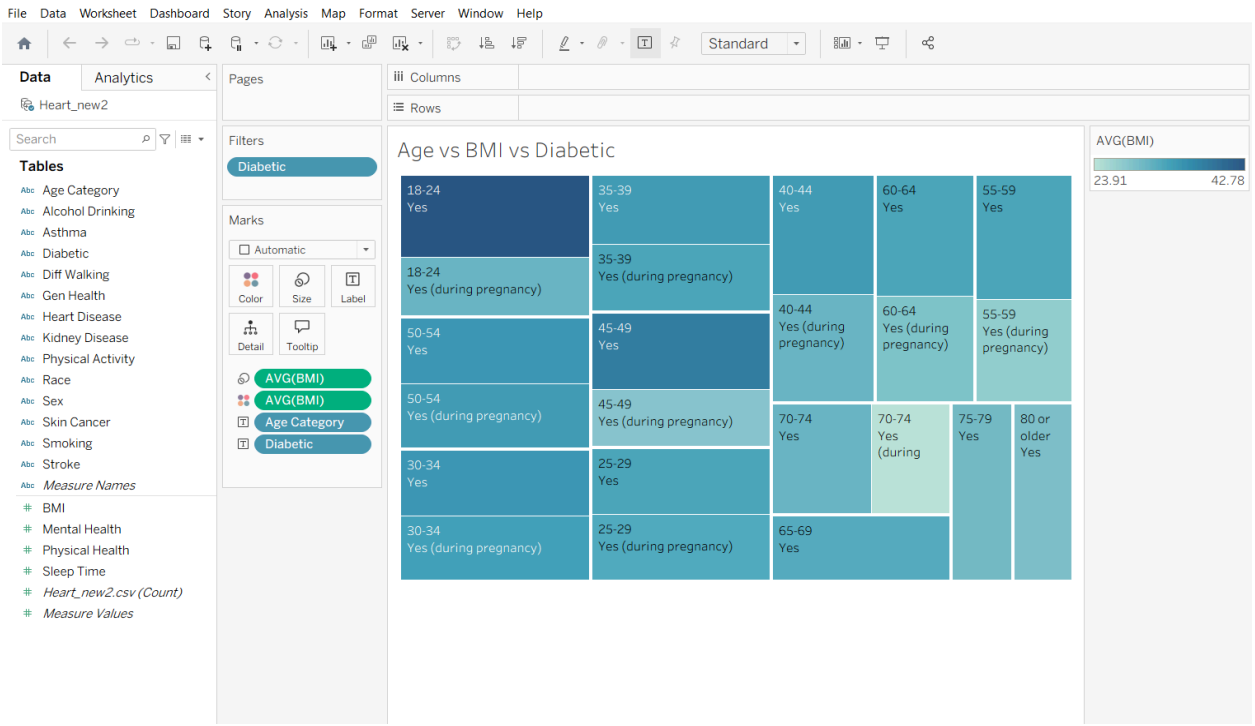
8) Move More Risk Less — Physical Activity vs Heart Disease

a) Drag and Drop PhysicalActivity into Columns section.
b) Drag and Drop CNT(HeartDisease) into the Angle shelf on the Marks card.
c) Add PhysicalActivity into Color from the Marks card.
d) Select Pie Chart / Donut Chart from Show Me panel.
e) Add CNT(HeartDisease) into Label to show Active: 3,231 and Inactive: 1,269.
f) To convert to Donut — add a second AVG(0) axis and set dual axis, then resize inner circle.



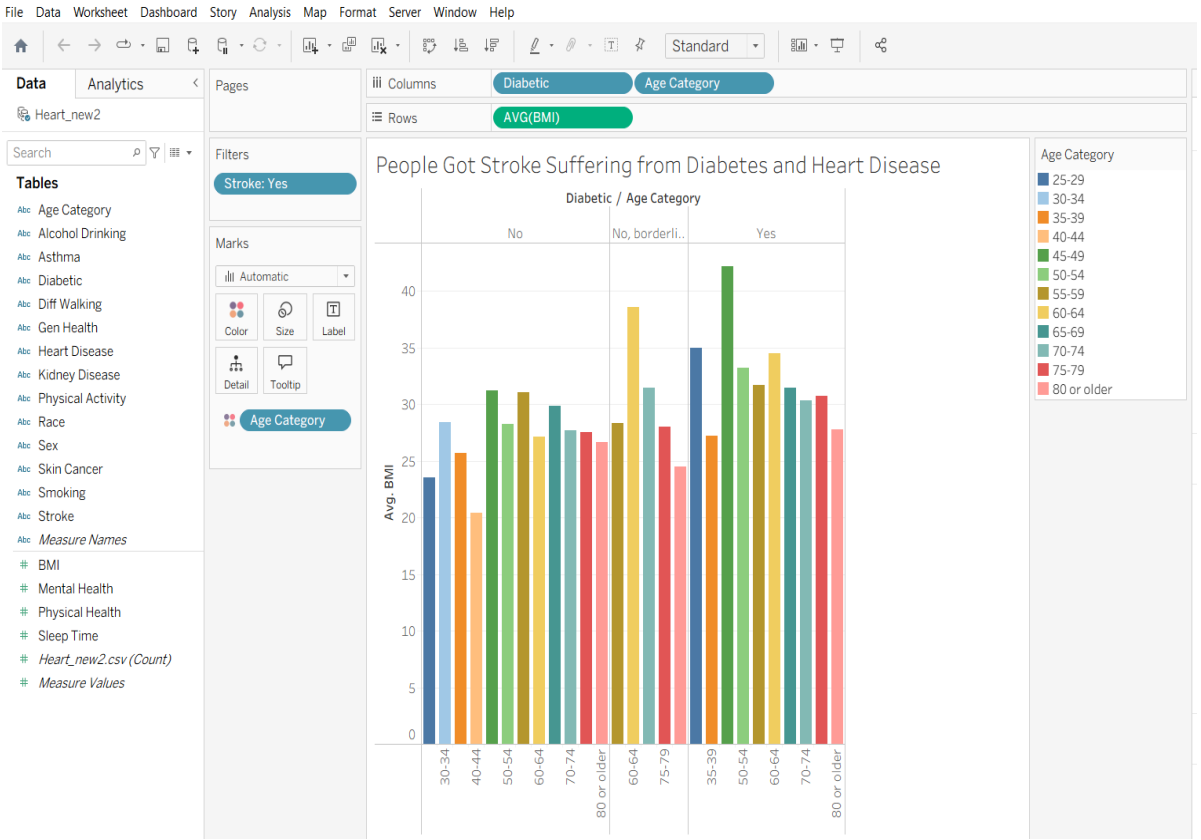
9) The Heavier the Years — Age and BMI vs Diabetic

a) Drag and Drop AgeCategory into Columns section.
b) Drag and Drop Diabetic into Rows section.
c) Add AVG(BMI) into Size from the Marks card.
d) Add Diabetic into Color from the Marks card.
e) Select Treemap from Show Me panel.
f) Add AgeCategory and AVG(BMI) into Label to display age group and average BMI value.



10) Triple Threat — Stroke, Heart Disease and Diabetic

a) Drag and Drop AgeCategory into Columns section.
b) Drag and Drop CNT(HeartDisease) into Rows section.
c) Add Diabetic into Color from the Marks card.
d) Filter Stroke to Yes to show only stroke patients.
e) Select Grouped Bar Chart from Show Me panel.
f) Add CNT(HeartDisease) into Label to display values on each bar.



4) Dashboard

A dashboard is a graphical user interface (GUI) that displays information and data in an organized, easy-to-read format. Dashboards are often used to provide real-time monitoring and analysis of data and are typically designed for a specific purpose or use case. Dashboards can be used in a variety of settings, such as business, finance, manufacturing, healthcare, and many other industries. They can be used to track key performance indicators (KPIs), monitor performance metrics, and display data in the form of charts, graphs, and tables.

❖ Creating Dashboard 1 — Heart Disease Analysis Dashboard

Step 1: Create a New Dashboard

a) At the bottom of Tableau, click the New Dashboard button (the icon next to the sheet tabs). b) A blank dashboard canvas will open.
c) The dashboard will be named Dashboard 1 by default.

Step 2: Set the Dashboard Size

a) On the left panel under Size, click the dropdown.
b) Select Fixed Size → choose Desktop Browser (1000 x 800) or select Automatic for responsive layout.

Step 3: Add Sheet 1 — Gender vs Heart Disease (Stacked Bar Chart)

a) From the left Sheets panel, drag Sheet 1 (Gender wise Heart Disease) onto the centre canvas.
b) Drop it in the top-left area of the dashboard.
c) On the right side of the sheet a downward arrow appears — click it and select Floating option if you want free positioning, or leave as Tiled for auto layout.
d) Resize the sheet by dragging its edges to fit the top-left quadrant.

Step 4: Add Sheet 2 — Age vs Heart Disease (Grouped Bar Chart)

a) From the left Sheets panel, drag Sheet 2 (Age wise Heart Disease) onto the canvas.
b) Drop it in the top-right area of the dashboard.
c) Resize to fill the top-right quadrant.

Step 5: Add Sheet 3 — Race-wise Heart Disease (Pie Chart)

a) From the left Sheets panel, drag Sheet 3 (Race wise Heart Disease) onto the canvas.
b) Drop it in the bottom-left area of the dashboard.
c) Resize to fill the bottom-left quadrant.

Step 6: Add Sheet 4 — Stroke vs Other Diseases (Bar Chart)

a) From the left Sheets panel, drag Sheet 4 (Stroke vs Other Diseases) onto the canvas.
b) Drop it in the bottom-right area of the dashboard.
c) Resize to fit the bottom-right quadrant.
d) You now have 4 visualizations arranged in a 2×2 grid layout.

Step 7: Add a Dashboard Title

a) At the bottom-left of the dashboard panel, check the box Show Dashboard Title.
b) Double-click the title area that appears at the top.
c) Type — Heart Disease Analysis Dashboard
d) Set font to Bold, size 16, color White.
e) Set background color to Red (#C0392B) by clicking Format → Dashboard → Title background.

Step 8: Add Interactive Filters

a) Click on any sheet inside the dashboard (e.g. Gender vs Heart Disease sheet).
b) Click the downward arrow on the top-right corner of that sheet.
c) Select Filters → check HeartDisease and Sex filters.
d) The filter widgets will appear on the right side of the dashboard.
e) Repeat for the Age sheet — add AgeCategory filter.
f) These filters will now control all connected sheets simultaneously.

Step 9: Format the Dashboard Background a)
Click Format from the top menu.
b) Select Dashboard from the Format panel.
c) Under Background set color to White or light grey #F4F4F4.
d) Under Dashboard Title set font size and background color.
e) To add borders to individual sheets — click each sheet → Format → Border → add outer border in grey or red.

Step 10: Add a Legend (if not auto-showing)
a) Click on the Gender vs Heart Disease chart inside the dashboard.
b) Click the downward arrow on the sheet.
c) Select Legends → check Color Legend.
d) The legend (Blue = No, Orange = Yes) will appear on the dashboard.
e) Drag and position the legend to a suitable corner.

Step 11: Publish the Dashboard to Tableau Public
a) Click File from the top menu → select Save to Tableau Public.
b) Sign in with your Tableau Public credentials if prompted.
c) Name the workbook — dashboard_HeartDisease or keep existing name.
d) Click Save — Tableau will upload and open the live link in your browser.
e) Your Dashboard 1 is now live at:
https://public.tableau.com/app/profile/priti.raje/viz/Heart_disease_analysis_17728097950570/Dashboard2?publish=yes

Creating Dashboard 2 — Heart Disease Analysis Dashboard

Step 1: Create New Dashboard
a) At the bottom of Tableau, click the New Dashboard button next to the sheet tabs.
b) A blank canvas will open named Dashboard 2.

Step 2: Set Size
a) On the left panel under Size → select Automatic or Desktop Browser (1000 x 800).

Step 3: Add 4 Sheets
a) From the left Sheets panel, drag and drop these 4 sheets onto the canvas:

Sheet 5 → General Health vs Heart Disease → Top Left
Sheet 6 → Physical Activity vs Heart Disease → Top Right
Sheet 7 → Age & BMI vs Diabetic → Bottom Left
Sheet 8 → Triple Threat Analysis → Bottom Right

b) Resize each sheet by dragging edges to fit into a 2×2 grid layout.

Step 4: Add Title

a) Check Show Dashboard Title at the bottom-left panel.

b) Double-click the title and type — Heart Disease Risk Factor Dashboard.

Step 5: Add Filters

a) Click any sheet → click the downward arrow on its top-right corner.

b) Select Filters → check HeartDisease, Diabetic, PhysicalActivity.

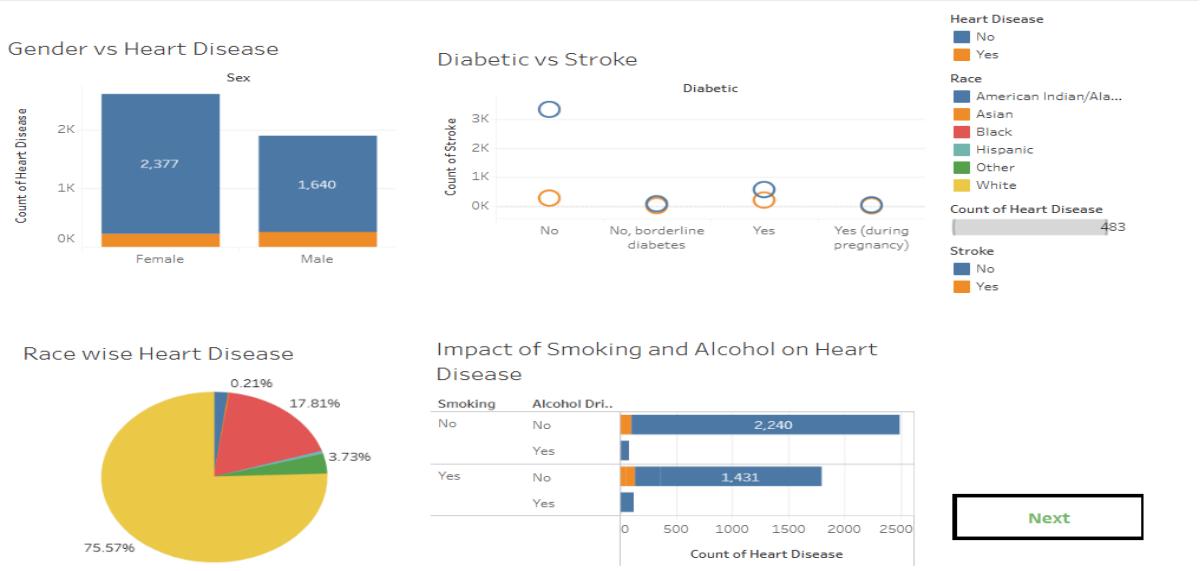
c) Filters will now control all 4 sheets simultaneously.

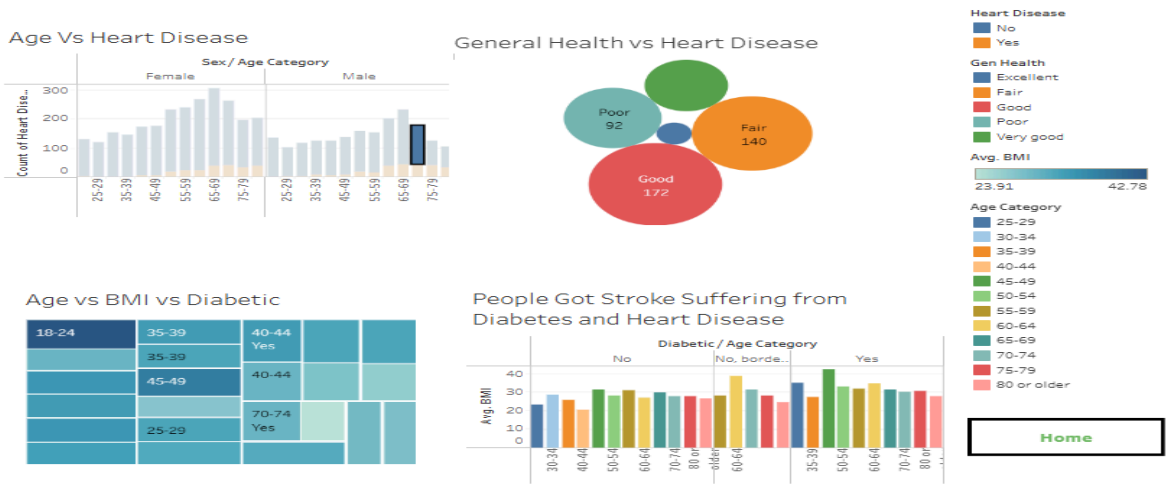
Step 6: Publish

a) Click File → Save to Tableau Public.

b) Sign in → Name the workbook → Click Save.

c) Dashboard goes live on your Tableau Public profile.





Story

A storyboard is a visual representation of a sequence of events, typically used in multimedia projects such as films, animations, advertisements, or presentations. It consists of a series of drawings or images arranged in a sequence, often accompanied by annotations or descriptions, to outline the flow of the story or concept.

Step 1: Create New Story
a) At the bottom of Tableau, click the New Story button next to the sheet tabs.
b) A blank story canvas will open named Story 1.

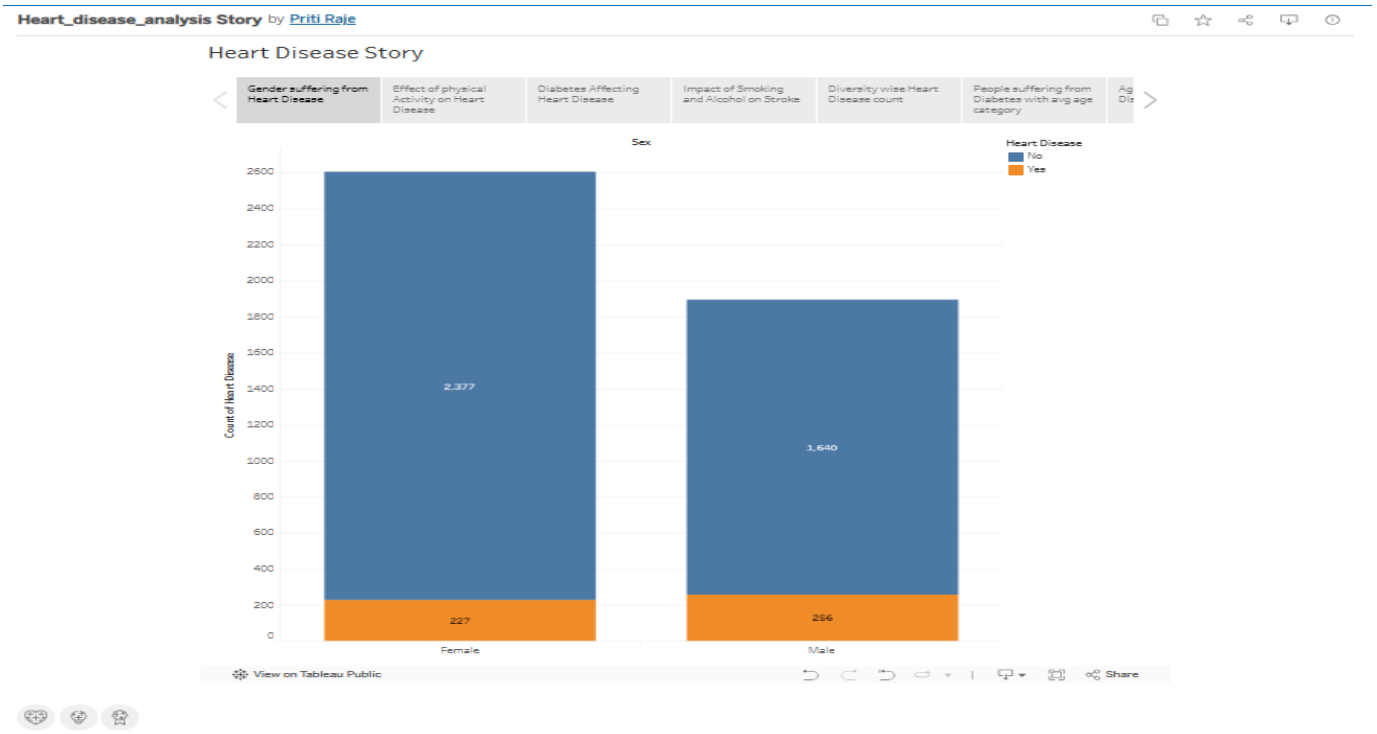
Step 2: Set Size
a) On the left panel under Size → select Automatic or Desktop Browser (1016 x 964).

Step 3: Add 10 Story Points
a) Click Blank on the left panel to create each new story point.
b) Drag and drop each sheet from the left Sheets panel onto the canvas one by one.
c) After dropping each sheet, click Add a Caption at the top and type the caption name.

Step 4: Add Story Title
a) Check Show Title at the bottom-left panel.
b) Double-click the title and type — Heart Diseases Story.

Step 5: Publish
a) Click File → Save to Tableau Public.
b) Sign in → Name the workbook → Click Save.
c) Story goes live on your Tableau Public profile.

https://public.tableau.com/app/profile/priti.raje/viz/Heart_disease_analysisStory/Story1?publish=yes



6) Web integration

Web integration of a Tableau involves embedding interactive visualizations into a website or web application. This allows users to explore data insights directly within a web interface, enhancing accessibility and engagement. It supports real-time updates, user filtering, and seamless navigation for a dynamic data storytelling experience.

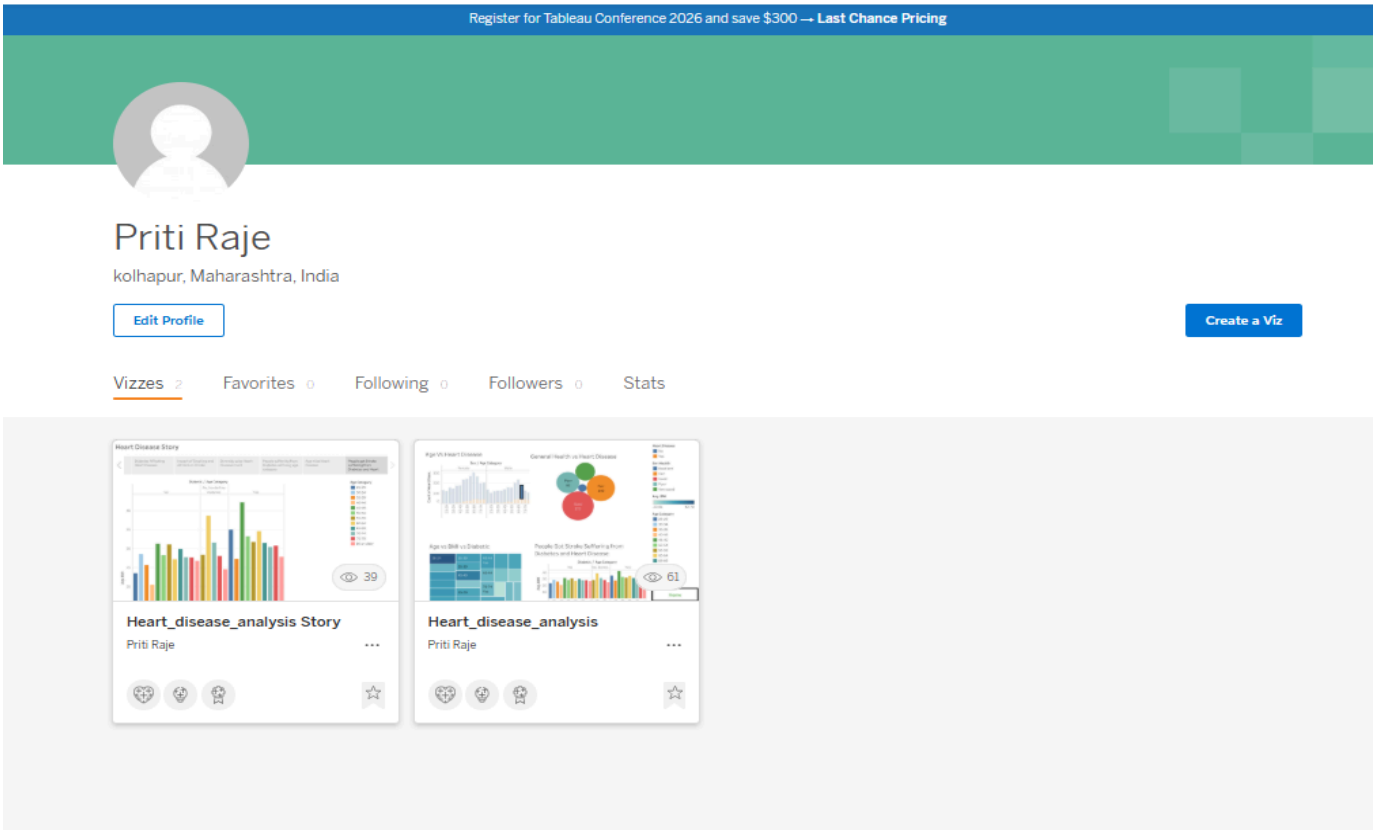
❖ Publishing

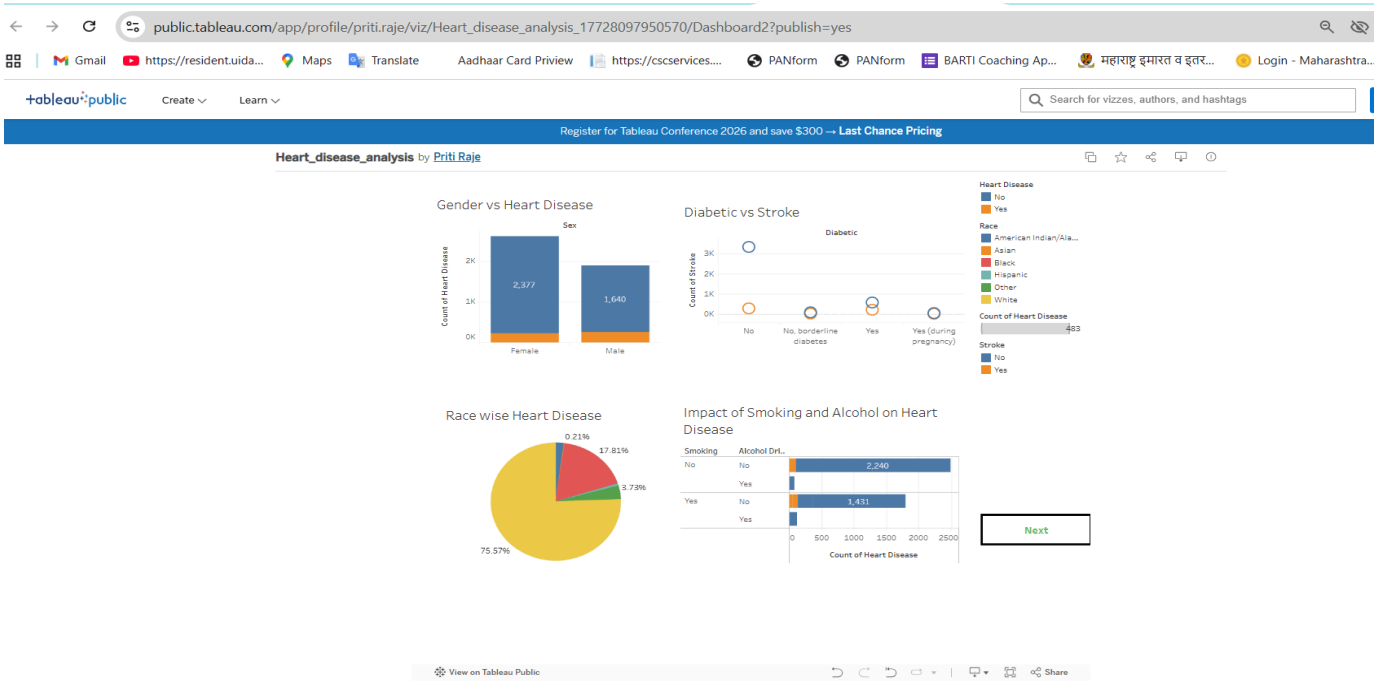
a) **Sign in to Tableau Public**

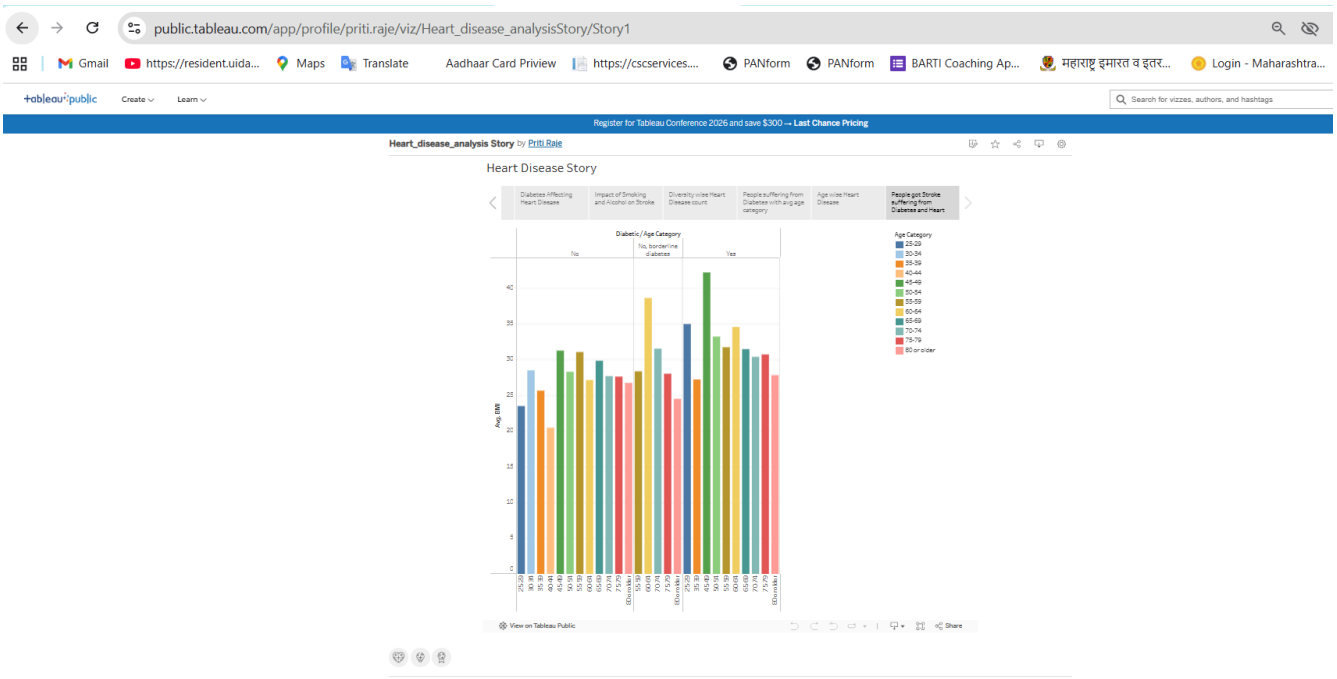
In Tableau Desktop, go to **File** > **Save to Tableau Public**.
If you're not already signed in, a login window will appear.
Enter your **Tableau Public credentials** or sign up if you don't have an account.

b) **Save and Publish**

- After logging in, you'll be prompted to **name your workbook**.
- Click **Save** – Tableau will upload the workbook to your Tableau Public profile.







❖ **Dashboard and Story embed with UI with Flask.**

In this activity, you will create a simple web application using Flask and embed a Tableau Dashboard or Story into the UI. This allows users to view and interact with Tableau visualizations directly through a custom web interface.

1) app.py

D:\Heart_Project\app.py

app.py × index.html ×

```python
import pickle
from flask import Flask, render_template, request

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

if __name__ == '__main__':
    app.run(debug=False)
```

2) Adding Dashboard and Story on HTML page.

a) Open Tableau Public profile and open Dashboard/Story.
b) Click on share option and copy the embed code.
c) Paste in HTML page

D:\Heart_Project\templates\index.html

app.py × index.html ×

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Heart Disease Analysis | Dashboard</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="s
    <style>
        body { background-color: #f8f9fa; font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sar
        .navbar { background-color: #004aad; }
        .hero-section { background: white; padding: 60px 0; border-bottom: 2px solid #eef0f2; }
        .viz-container { background: white; padding: 20px; border-radius: 10px; box-shadow: 0 4
        .section-title { color: #004aad; font-weight: bold; margin-bottom: 25px; border-left: 5
        footer { background: #333; color: white; padding: 20px 0; margin-top: 50px; }
    </style>
</head>
<body>

    <nav class="navbar navbar-expand-lg navbar-dark shadow-sm">
        <div class="container">
            <a class="navbar-brand" href="#"><strong>Welcome</strong> | Heart Disease Analysis<
            <div class="collapse navbar-collapse justify-content-end">
                <ul class="navbar-nav">

                    <li class="nav-item"><a class="nav-link" href="#Home">Home</a></li>
                    <li class="nav-item"><a class="nav-link" href="#dashboard">Dashboard</a></l
                    <li class="nav-item"><a class="nav-link" href="#story">Story</a></li>
                    <li class="nav-item"><a class="nav-link btn btn-primary btn-sm ms-lg-3 text
                </ul>
            </div>
        </div>
    </nav>

    <section class="hero-section text-center">
        <div class="container">
            <h1 class="display-4 fw-bold">Heart Disease Analysis</h1>
            <p class="lead text-muted">Using Data Science and Flask to drive healthcare insight
        </div>
    </section>
```

d) Run the app.py and copy the port number and paste it on Browser.

Console 1/A ×

```
Python 3.12.11 | packaged by conda-forge | (main, Jun  4 2025, 14:29:09) [MSC v.1943 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 9.10.0 -- An enhanced Interactive Python. Type '?' for help.

In [1]: %runfile D:/Heart_Project/app.py --wdir
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI
server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
```

Welcome | Heart Disease Analysis

Home Dashboard Story Get Started

Heart Disease Analysis

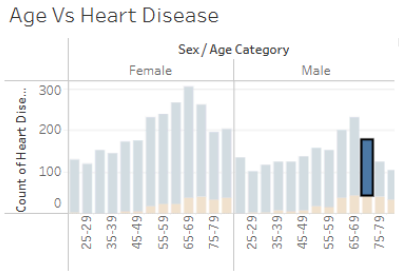
Using Data Science and Flask to drive healthcare insights.

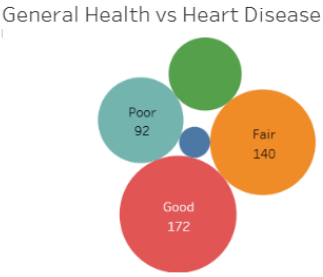
Heart Disease Analysis

Heart Disease is a common health condition that affects the heart and blood vessels. It can be caused by factors such as high blood pressure, high cholesterol, smoking, diabetes, and an unhealthy lifestyle. Common symptoms include chest pain, shortness of breath, and fatigue. Maintaining a healthy diet, exercising regularly, and having regular health check-ups can help reduce the risk of heart disease.Explore factors like Smoking, Alcohol, and Diabetes impact on Heart Health. Common symptoms include chest pain, shortness of breath, fatigue, dizziness, and irregular heartbeat. In some cases, heart disease may develop slowly without noticeable symptoms, which makes regular health check-ups important. Adopting a healthy lifestyle can help prevent heart disease. Eating a balanced diet, exercising regularly, avoiding smoking, maintaining a healthy weight, and managing stress are important steps to keep the heart healthy. Early diagnosis and proper treatment can significantly reduce the risk and improve overall heart health. ❤️

Interactive Analysis Dashboard

Explore factors like Smoking, Alcohol, and Diabetes impact on Heart Health.



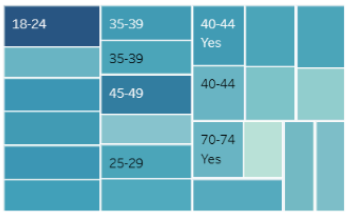


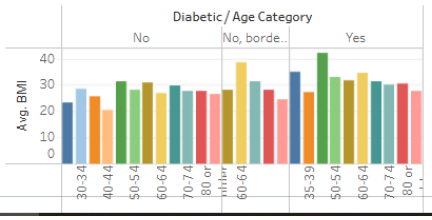
Heart Disease
- No
- Yes

Gen Health
- Excellent
- Fair
- Good
- Poor
- Very good

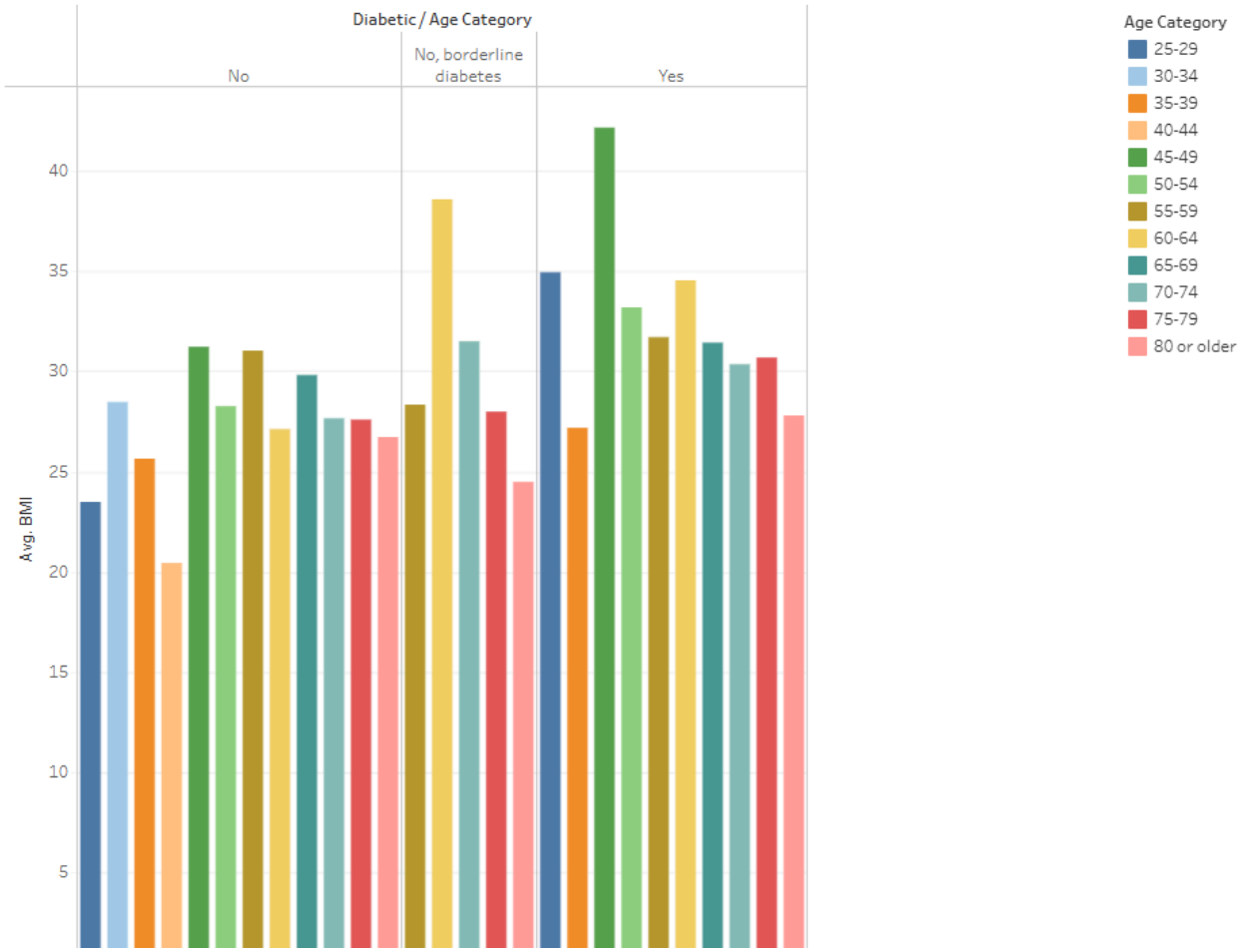
Avg. BMI
23.91 42.78

Age Category
- 25-29
- 30-34
- 35-39
- 40-44
- 45-49
- 50-54
- 55-59
- 60-64
- 65-69
- 70-74
- 75-79
- 80 or older





Home

Heart Disease Story

Diabetes Affecting Heart Disease | Impact of Smoking and Alcohol on Stroke | Diversity wise Heart Disease count | People suffering from Diabetes with avg age category | Age wise Heart Disease | People got Stroke suffering from Diabetes and Heart